Correlation between individual differences in striatal dopamine and in visual consciousness

Filip Van Opstal[1,*], Nick Van Laeken[2], Tom Verguts[1], Jean-Philippe van Dijck[1], Filip De Vos[2], Ingeborg Goethals[3], and Wim Fias[1]

A widely held view on consciousness is that it is related to the 'broadcasting' of sensory information to the whole brain [1–3]. Despite the fact that there is general support for this view, it remains unclear how exactly this broadcasting is established. It has been proposed [2,3] that thalamocortical circuits are an important mediator of such broadcasting, but empirical support for this claim is lacking. In the present study, we investigated this hypothesis by exploiting the well-established, but in this context neglected, fact that thalamocortical connectivity is modulated by dopaminergic activity in the striatum [4]. We used positron emission tomography (PET) to measure individual differences in striatal dopamine (DA) level and we correlated this with individual differences in visual consciousness. Our results show that visual awareness is related to the concentration of endogenous DA or DA receptors in striatal areas, supporting the importance of dopaminergic signalling in visual consciousness.

The thalamus is an important relay station for propagating sensory information: it relays information from ascending pathways from subcortical and peripheral areas to cortex, and also modulates cortico-cortical communication [5]. Thalamocortical connectivity therefore plays a major role in making sensory information widely available to the brain, a theoretical prerequisite for consciousness [1]. Effective thalamocortical connectivity is known to be modulated by dopaminergic activity in the striatum that acts on a closed cortico-basal ganglia-thalamocortical (C-BG-Th-C) loop [4]. Although this fact has been largely neglected in models of consciousness, it could be that striatal DA is involved in broadcasting sensory information to the brain, and hence to visual consciousness. We measured striatal DA by the binding potential of the radioligand [11C] raclopride to DA D2 receptors, which is known to be sensitive to levels of endogenous striatal DA. A voxel-wise binding potential was obtained by coregistering the PET image with the individual's T1 weighted MR image (MPRAGE; Figure 1C and the online Supplemental Information for details). Objective and subjective visibility was measured by a behavioral backward masking experiment in which participants first had to classify a target number as smaller or larger than five, and then indicate the visibility of the target number (*seen* or *unseen*) [6,7]. The visibility of the target stimulus was manipulated by changing the stimulus onset asynchrony (SOA) between the target stimulus and a subsequent masking stimulus (Figure S1). The objective and subjective visibility was defined by the accuracy of classifying the target number and by the responses to the visibility question, respectively (see Supplemental Information).

Individual differences in objective and subjective visibility were highest in the SOA 16 ms and 33 ms conditions (Figures 1A,B and Table S1), with objective and subjective visibility, averaged across these two SOA conditions, being correlated (R^2 = 0.61, P = 0.002). Analysis on the voxel-wise binding potentials with the averaged objective visibility measure as a covariate revealed a significant positive relation in right putamen (peak at 25, –9, –4; T = 10.03; cluster size: 1869 voxels; P < 0.001). Subjective performance also revealed a positive relation with binding potential in right putamen (peak at 24, –10, –6; T = 4.96; cluster size: 177 voxels; P = 0.054). No negative correlation between objective or subjective performance and binding potential was found. An overlay of the clusters for objective and subjective visibility shows a large overlap in right putamen (Figure 1D). Extraction of the average binding potential in the clusters found for subjective and objective visibility shows a linear increase with visibility threshold in right putamen (Figure 1E). No significant clusters were found when the false alarm rate of the no prime condition was used as a covariate.

Additional analyses were performed to investigate the reliability and specificity of our results. Reliability was investigated by means of a jackknife procedure and by using other visibility measures (for example, for SOA 16 and 33 separately, or across all SOA's). The results always revealed a similar cluster in right putamen (see Supplemental Information). The specificity was demonstrated by showing that the results were not caused by other task-related variables, such as processing speed or other general information processing characteristics (see Supplemental Information).

These results show the involvement of striatal DA in visual consciousness. Although these results are only correlational and a causal relation will need to be established in future work, here it is demonstrated that D2 binding potential in the right putamen is positively related to both subjective and objective visibility. In line with earlier observations of a functional connectivity between right putamen and both the frontal and parietal cortex and the thalamus [8], the results of the present study suggest that previous models implicating effective thalamocortical connectivity to consciousness should be extended by adding a dopaminergic mesocircuit that modulates this connectivity through a C-BG-Th-C loop. Such an extended model is consistent with the recent observation of an increased threshold in schizophrenic patients [6]. Abnormal dopaminergic transmission in these patients would restrain the broadcasting of sensory information to the cortex and hence increase the consciousness threshold. Because of the close relation between striatal DA and working memory [9], the model can also

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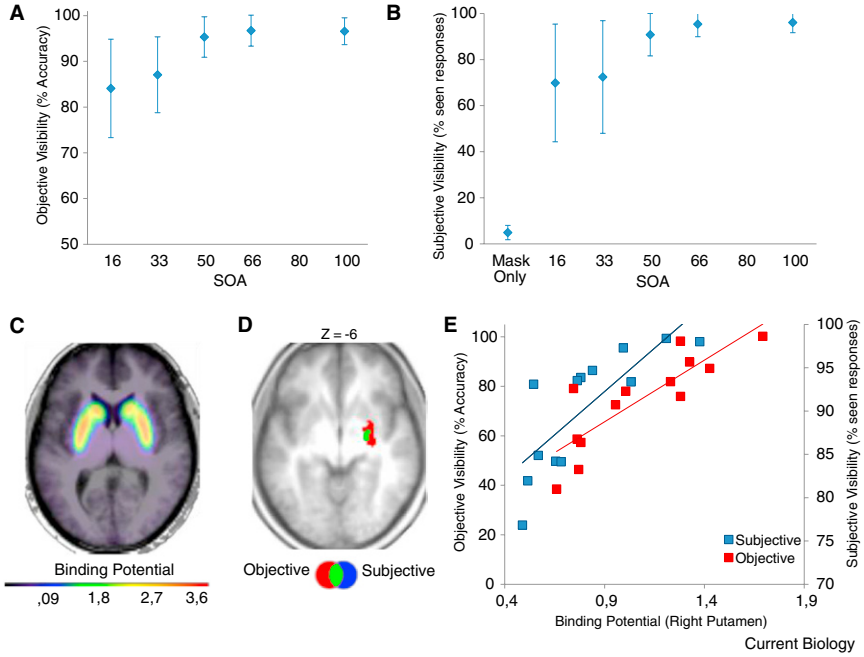


Figure 1. Behavioral and PET results.
(A) Average objective visibility calculated from the responses on the target classification task. Error bars denote the standard deviation. Individual differences in objective visibility are highest in the SOA 16 and SOA 33 ms conditions (see also Table S1). (B) Average subjective visibility calculated from the responses to the visibility question. Responses to the no prime condition are indicative of an individual's response bias (conservativeness to report the presence of a stimulus). Error bars denote the standard deviation. Individual differences in objective visibility are highest in the SOA 16 and SOA 33 ms conditions (see also Table S1). (C) Binding potential averaged across all participants, overlaid on the average anatomical image. (D) Binding potential had a positive correlation with both subjective and objective visibility in right putamen. (E) Extraction of the average binding potential in right putamen clusters illustrates the strong positive correlation between binding potential and objective and subjective visibility.

explain why an increased visibility threshold is observed when working memory is loaded [7].

The C-BG-Th-C loop proposed here for consciousness is similar to the mechanism hypothesized to support updating and maintaining of working memory information [9]. Maintaining information in working memory is subserved by striatal DA activity that reduces thalamocortical communication to prevent overwriting working memory with new information. It could thus be speculated that the reduced effective connectivity caused by working memory maintenance also prevents new information to become globally available to the brain. This might possibly explain why visibility threshold increases under high working memory load [7], but future research is needed to test this hypothesis explicitly.

The extended model is also reminiscent of a recent 'mesocircuit' model for the recovery of consciousness after brain injury [10]. According to this model, downregulation of the C-BG-Th-C loop is predicted to have a broad impact on the global dynamics of the thalamo-cortical connectivity which results in changes in consciousness states. Our observation that implicates striatal DA in visual awareness, possibly by regulating thalamo-cortical connectivity through the C-BG-Th-C loop, could indicate a common neural mechanism for both intransitive consciousness (a *state* of consciousness, for example coma or sleep) and transitive consciousness (conscious access of a piece of information, for example a visual stimulus).

Supplemental Information

Supplemental Information includes data, one table, experimental procedures, and one figure and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.001.

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[1]Department of Experimental Psychology, Ghent University, Belgium. [2]Department of Pharmaceutical Analysis, Ghent University, Belgium. [3]Department of Nuclear Medicine, Ghent University, Belgium.
*E-mail: Filip.VanOpstal@ugent.be

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